

Research Interests

Security, Cloud-Native, Distributed Networked System, Anomaly Detection, eBPF-based Provenance

Education

Pusan National University

M.S. in Computer Engineering

Busan, Republic of Korea

Mar. 2024 – Feb. 2026

Pusan National University

B.S. in Computer Sciences and Engineering

Busan, Republic of Korea

Mar. 2018 – Feb. 2024

Served mandatory military service

May. 2020 – Nov. 2021

Employment

Zero Trust Cloud Security Research Center

Researcher

Busan, Republic of Korea

Mar. 2026 – Aug. 2026

ERUTY

Software Engineer

Busan, Republic of Korea

Dec. 2022 – Mar. 2023

- Contributed as an early-stage developer to backend development of a 3D graphic design marketplace platform.

Publications

- Hybrid Session-Based UEBA for AWS Environment**

*Doo-Hwan Choi, **Geonwoo Yoon**, Jae-Seok Kim, Yoon-Ho Choi*

The 17th International Conference on Ubiquitous and Future Networks (ICUFN), 2026.

- Container-Specific Service Mesh-based System for Mitigating Lateral Movement Attacks**

***Geonwoo Yoon**, Jinmyeong Shin, Jae-Seok Kim, Seunghyuk Kim, Jaeyoung Jeong, Yoon-Ho Choi*

IEEE Transactions on Cloud Computing, 2026. [Paper]

- Lightweight Service Mesh for Intrusion Detection using KD-CNN in Cloud-Native Environments**

***Geonwoo Yoon**, Jae-Seok Kim, Seunghyuk Kim, Jaeyoung Jeong, Millati Pratiwi, Yoon-Ho Choi*

ACM Cloud Computing Security Workshop (CCSW in conjunction with the ACM CCS), 2025. [Paper] [Video]

- Automated Cybersecurity Risk Assessment and Visualization Framework for Resilient Cloud IT Asset Management**

*Joonho Seo, Jinmyeong Shin, **Geonwoo Yoon**, Jae-Seok Kim, Yoon-Ho Choi*

ACM/SIGAPP Symposium on Applied Computing (SAC), 2025. [Paper]

Under Review

- Agentic Detection-Aware Penetration Testing in Monitored Environments**

*Seunghyuk Kim, **Geonwoo Yoon**, Yoon-Ho Choi*

Under review (double-blinded)

- A Contrasive Learning-Driven Probabilistic Model Scheduling Framework for Mitigating MTDDoS Attacks in Cloud-Native AI Services**

*Jae-Seok Kim, Seunghyuk Kim, **Geonwoo Yoon**, Joonho Seo, Yoon-Ho Choi*

Under review (double-blinded)

- SCALE: Risk-aware Distributed System Call Audit Framework for Cloud-Native Environments**

***Geonwoo Yoon**, Seunghyuk Kim, Jaeyoung Jeong, Jae-Seok Kim, Yoon-Ho Choi*

Under review (double-blinded)

- **Context-Preserving Adversarial System Call Injection for Cloud Anomaly Detection Evasion**

Jae-Seok Kim, Seunghyuk Kim, **Geonwoo Yoon**, Joonho Seo, Yoon-Ho Choi

Under review (double-blinded)

Projects

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| Online Learning-based Service Mesh for Lateral Movement Attacks | Developed by CSRC |
| <i>Project Manager & Research Engineer</i> | <i>Feb. 2026 – Apr. 2026</i> |
| <ul style="list-style-type: none"> • Planned development schedules, roles, and scope as a project manager. • Designed and implemented a proxy-based data plane leveraging iptables for traffic control and anomaly detection. • Designed and implemented a control plane for model training and deployment for online learning. | |
| Enhancing VPD Control Web Framework (SCOPE) | In collaboration with LG Electronics |
| <i>Developer</i> | <i>Aug. 2025 – Dec. 2025</i> |
| <ul style="list-style-type: none"> • Optimized SCOPE framework architecture to improve scalability, stability, and code quality. • Enhanced security and accessibility through testing and user-centered UI/UX improvements. • Improved maintainability with technical documentation and framework refinement. | |
| PNU Zero Trust CSRC CASB (Cloud Access Security Broker) [Web] | Developed by S3Lab |
| <i>Research Engineer</i> | <i>Jul. 2024 – May. 2025</i> |
| <ul style="list-style-type: none"> • Designed and developed a CASB to address multi-cloud access control, data leakage, and anomaly behavior. • Implemented CASB Client (VPN Client), CASB Server (Web, Logging, and Anomaly Detection Server), and CASB Agent (VPN Client, Logging). • Developed synchronization logic for AWS and Azure clouds, and monitoring functionality. | |
| Industrial Coating Film Defect Detection and Monitoring System | In collaboration with GBLIGHT |
| <i>Research Engineer</i> | <i>Jul. 2024 – Dec. 2024</i> |
| <ul style="list-style-type: none"> • Implemented logic for the analysis server to control the PLC module via OPC UA communication. • Preprocessed coating film datasets, trained detection models using YOLOv5, and deployed them to production. • Developed the analysis server for real-time monitoring and reporting. | |
| Web-based Psychoeducational Platform [Web] | In collaboration with the Dept. Psychology, PNU |
| <i>Project Manager & Developer</i> | <i>May. 2024 – Nov. 2024</i> |
| <ul style="list-style-type: none"> • Planned development schedules, roles, and scope as a project manager. • Set up the initial development environment and contributed to frontend, backend, and database development. • Deployed the platform to production and performed ongoing maintenance. | |
| PNU Cloud RMF (Risk Management Framework).v1 | Developed by S3Lab |
| <i>Research Engineer</i> | <i>Dec. 2023 – May. 2024</i> |
| <ul style="list-style-type: none"> • Analyzed NIST SP 800-37, SP 800-53, and SP 800-30 standards and devised strategies for cloud application. • Designed and developed the <i>Select</i>, <i>Implement</i>, and <i>Assess</i> steps of the RMF according to NIST standards. • Implemented automated report generation functionality. | |

Teaching Experience

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| Operating Systems | Pusan National University |
| <i>Teaching Assistant</i> | <i>1st Semester 2024</i> |

Honors and Awards

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| Spear-phishing response training mini-challenge | Dec. 2023 |
| <ul style="list-style-type: none"> • Awarded Third Place in the spear-phishing response training mini-challenge, hosted by the Ulsan Information Security Center and Korea Internet & Security Agency (KISA). | |
| 4th PATH-HACK Makerthon | Feb. 2023 |
| <ul style="list-style-type: none"> • Awarded First Place in the PATH-HACK Makerthon for developing an integrated software–hardware prototype, hosted by pathfinder & Korean Institute of Startup and Entrepreneurship Development. | |
| Academic Excellence Scholarship | 1st & 2nd Semester. 2022 |
| National Science & Technology Scholarship | 1st Semester 2018 |
| <ul style="list-style-type: none"> • Merit-based national STEM scholarship awarded for outstanding scores in math and science on the Korean CSAT. | |

Patents

- **Geonwoo Yoon**, Yoon-Ho Choi, "RISK-BASED DIFFERENTIATED DISTRIBUTED SYSTEM CALL COLLECTION FRAMEWORK DEVICE AND ITS OPERATION METHOD IN A CLOUD-NATIVE ENVIRONMENT". Korean Patent Application No. 10-2026-0068229 (Pending)
- **Geonwoo Yoon**, Yoon-Ho Choi, "DISPATCHER-BASED DISTRIBUTED SYSTEM CALL COLLECTION DEVICE AND METHOD FOR CLOUD-NATIVE ENVIRONMENTS". Korean Patent Application No. 10-2026-0068266 (Pending)
- **Geonwoo Yoon**, Yoon-Ho Choi, "METHOD AND SYSTEM FOR DETECTING ABNORMAL BEHAVIOR TO PREVENT LATERAL MOVEMENT ATTACKS IN KUBERNETES ENVIRONMENT". Korean Patent Application No. 10-2025-0067413 (Pending)
- **Geonwoo Yoon**, Yoon-Ho Choi, "ZERO-COPY BASED MAIN CONTAINER'S SYSTEM CALL COLLECTION DEVICE AND METHOD FOR USING EBPF IN KUBERNETES SECONDARY CONTAINER". Korean Patent Application No. 10-2025-0067396 (Pending)
- Jae-Seok Kim, **Geonwoo Yoon**, Yoon-Ho Choi, "EQUIPMENT CONTROL BASED VISION METHOD AND SYSTEM FOR AI-BASED DEFECT DETECTION AND DATA COLLECTION IN HIGH-SPEED AND WIDE-AREA REFLECTIVE FILMS". Korean Patent Application No. 10-2024-0133547 (Pending)

Skills

Languages	C/C++, Python, JavaScript
Libraries	Pytorch, TensorFlow, scikit-learn, React, FastAPI
Tools	Docker, Kubernetes, eBPF, PostgreSQL, pgAdmin